

Robert A. Frenken

frenken.2@osu.edu · Columbus, OH · Website  GitHub ·  Hugging Face ·  LinkedIn ·  Scholar

EDUCATION

The Ohio State University

Columbus, OH

Ph.D. in Industrial & Systems Engineering (Expected May 2027)
M.S. in Industrial & Systems Engineering
B.S. in Industrial & Systems Engineering

August 2021 – Present
May 2024
May 2021

MS GPA: 3.83 / BS GPA: 3.63

Relevant Coursework: Machine Learning of Large Language Models, Learning-Based Controls, Nonconvex Optimization for ML, High Performance Computing, Computer Vision, Reinforcement Learning, Human-Centered ML, Data Mining

PROFESSIONAL EXPERIENCE

Critical Infrastructure Analysis Graduate Intern | Lawrence Livermore National Laboratory

Summer 2026

Cyber and Critical Infrastructure Summer Institute. (*Details forthcoming.*)

Graduate Research Assistant | Ohio State University

2021 – Present

Center for Automotive Research (2024 – Present)

- Developing KD-GAT, a multi-stage intrusion detection system for CAN bus networks using knowledge distillation, variational graph autoencoders, and graph attention networks. Trained on the Ohio Supercomputer Center (SLURM) with Ray-based distributed hyperparameter tuning.
- Created a federated learning testing bed configurable to be either centralized or decentralized, with the ability to change dataset, model, graph topology, and aggregation strategy.

Spine Research Institute (2021 – 2024)

- Created multiple supervised and unsupervised machine learning models to classify lower back pain from wearable data of over 1000 participants. The supervised ensemble CNN model achieved 90% accuracy.

Data Science Intern | Ford Motor Co. | Columbus, OH (Remote)

2020

- Leveraged HPC tools (Spark, Hadoop, HIVE, SQL) and Python to query, merge, and aggregate trip data.
- Developed a decision tree model analyzing the correlation between oil life efficiency and warranty repair rates.

Machine Learning Research Assistant | Ohio State University

2020 – 2021

- Built web-scraping program to automate extraction of weekly FanDuel player metrics. Conducted clustering and sentiment analysis to inform lineup decisions.

Quality Systems Data Analyst Co-op | Momentive

2019

Built Python/Excel data pipeline for HVAC metrics, production yields, and machine reliability tracking.

Project Engineering Co-op | DuPont

2018

Executed 12 capital projects (\$1M+ total), averaging 10% under budget. Installed roller lifts, pallet yards, and pumps.

PUBLICATIONS

- **R Frenken**, SG Bhatti, H Zhang, Q Ahmed (2025) “KD-GAT: Combining Knowledge Distillation and Graph Attention Transformer for a Controller Area Network Intrusion Detection System” *IEEE ITSC 2025* (to appear).
- **R Frenken**, SG Bhatti, H Zhang, Q Ahmed (2025) “Multi-Stage Knowledge-Distilled VGAE and GAT for Robust Controller-Area-Network Intrusion Detection” (preprint).

- WM Wu, H Zhang, **R Frenken**, Z Li, SG Bhatti, Q Ahmed (2026) “CWD–SWG–IDS: Complementary Fusion of Temporal Context and Structural Graphs for CAN Intrusion Detection via Transformer” *IEEE ITSC 2026* (under review).

PROJECTS

Ohio Campaign Finance Dashboard

2026 – Present

- Interactive data visualization platform exploring political contributions and spending across Ohio legislative districts
- Built with D3.js, DuckDB-WASM, and Observable Framework — 6 pages, 18+ chart types (choropleths, Sankey diagrams, network graphs), real-time filtering across 28K+ records
- LLM-assisted donor and expenditure classification pipeline

Center for AI Safety (CAIS) 2025 Capstone Project

2025

- C Leet, **R Frenken**. Tacit Coordination: Coordination without Communication
- Empirical investigation of LLM behavior in game-theoretic scenarios (prisoner’s dilemma, battle-of-the-sexes, duopoly markets)

Mobility Systems Lab Resources | CAR Mobility Systems Lab

2026 – Present

- Creator and maintainer of the lab’s technical knowledge base and HPC resource usage dashboard
- Reusable Quarto research templates — covers HPC workflows, PyTorch/PyG setup, SLURM, and ML best practices

AgentBeats CRMarena Competition | Berkeley RDI

2026

- Built an A2A protocol agent for a CRM business process benchmark — schema drift reversal, category-specialized routing across 22 task types, single-turn optimization for token efficiency

TECHNICAL STACK

Applications	D3.js · Observable · ECharts · HF Spaces · Cloudflare · GitHub Pages
ML Pipeline	PyTorch · PyTorch Geometric · Ray · MLflow · scikit-learn · Knowledge Distillation · Federated Learning
Data	DuckDB · SQL · Pandas · GeoPandas · Spark · NumPy
Infrastructure	SLURM · OSC · Docker · Apptainer · Git · GitHub Actions · Linux
Languages	Python · Java · SQL · JavaScript · MATLAB · HTML/CSS

SERVICE & MENTORSHIP

Reviewer: ITSC 2025, AAAI 2025

Mentorship:

- Hanqin Zhang — B.S. Computer Science ‘25, Ohio State · now M.S. Georgia Tech
- Drew Ralston — B.S. Computer Science ‘28 (expected), Ohio State
- Chunyu Gu — B.S. Mathematics ‘28 (expected), Ohio State